# From Reactive to Predictive: A Federated Graph-Attention SAC Framework for Schedule-Driven WiFi Optimization in High-Density Transit Venues

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**Abstract**

High-density transit venues—such as airport terminals, railway stations, and bus interchanges—experience severe, schedule-driven congestion where passenger crowds gather predictably around arrival and departure events. Traditional WLAN controllers operate reactively, balancing radio resources only after congestion causes packet loss and SLA violations. In this work, we propose **STAG-FedSAC**, an end-to-end framework for proactive, schedule-driven WiFi radio resource management. STAG-FedSAC integrates three coordinated modules: (i) **ST-GCAT**, a schedule-conditioned spatio-temporal graph attention Transformer that predicts AP-level load up to 30 minutes ahead by cross-attending multi-step transit timetables with spatial AP interference topologies; (ii) multi-agent **Soft Actor-Critic (SAC)** agents per AP that perform proactive, multi-objective control over transmit power, channel selection, and quantized 802.11e EDCA airtime quotas under a Lagrangian airtime constraint; and (iii) **HierFed-KD**, a hierarchical federated distilla-

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tion protocol that combines intra-zone parameter averaging with inter-zone proximal knowledge distillation over a public reference set, preserving multi-operator data privacy while drastically cutting communication overhead.

To ensure physical and protocol realism, our evaluation incorporates IEEE 802.11 MAC contention based on Bianchi's saturation model, channel-switching outage penalties, and hardware-aligned EDCA airtime budgeting. Across multi-seed evaluations, STAG-FedSAC achieves superior performance across all operational metrics: it attains a mean user throughput of 1.32 Mbps, reduces average user latency to 109.9 ms (an 11.8% reduction compared to the static incumbent plan SSF at 124.6 ms), and boosts QoS satisfaction to 0.689, with excellent energy efficiency (7.47 Mbps/W) and stable channel management (0.15 switches/AP/step). Relative to learned baselines, STAG-FedSAC outperforms FedDDPG and LSTM-DRL by +56.9% and +68.5% in reward and reduces latency by 40.5% and 44.7%, respectively, avoiding the severe channel-switching thrashing that undermines non-entropy-regularized policies. Furthermore, ST-GCAT cuts load forecast RMSE by 32% overall and 39.5% in critical surge windows relative to persistence, outperforming schedule-unaware models by over 23%. HierFed-KD reduces inter-zone wide-area communication by 34× relative to weight-based hierarchical federation and 1711× relative to flat FedAvg. Extensive ablation and sensitivity analyses provide concrete design principles for schedule-aware wireless reinforcement learning.

*Keywords:* WiFi resource management, Spatio-temporal graph attention, Soft Actor-Critic, Federated learning, Knowledge distillation, Transit venue networks, Deep reinforcement learning, Schedule-aware prediction

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## 1. Introduction

Transit venues—airports, railway stations, and bus terminals—are among the most demanding environments for WiFi. At peak hours a single departure lounge can host hundreds of users streaming video, making calls, and uploading content over a handful of access points (APs) [**?** ]. Unlike enterprise offices or university campuses, transit venues have a critical feature that a network controller can exploit: a substantial fraction of the congestion is *announced in advance*. Departure and arrival timetables indicate when and where passenger crowds will gather, with a temporal precision rarely available in other wireless settings. Commercial WLAN controllers, however, remain almost entirely *reactive*: they detect congestion after users experience degraded service and respond with local load-balancing heuristics that ignore future schedule information [**?** ].

In this paper, we investigate how schedule information can be systematically integrated into proactive radio resource optimization. We organize the investigation around four fundamental research challenges:

1. **Schedule unawareness**: conventional load predictors treat AP load as a purely historical time series and discard timetable metadata that provides crucial lead time before an impending surge [**?** ].
2. **Reactive radio resource control**: heuristic controllers (e.g., strongest signal first or least loaded first) and conventional deep reinforcement learning (DRL) agents trained without future look-ahead cannot preconfigure channel assignments, transmit powers, and airtime allocations before crowds physically arrive [**?** ].

3. **Multi-class QoS alignment**: traffic in transit hubs is heterogeneous: VoIP requires ultra-low latency, video demands high throughput, and best-effort traffic tolerates delay. This necessitates a hybrid, multi-objective action space (continuous transmit power, discrete channel choice, and per-class airtime budgeting) that standard continuous DRL architectures struggle to coordinate stably [**?** ].
4. **Decentralized learning across multi-operator administrative domains**: in large transit terminals, distinct zones often belong to different administrative operators (e.g., airport authority, airline lounges, retail tenants) who cannot pool sensitive per-user association and traffic data. Isolated per-AP learners fail to exploit broader spatial correlations, while monolithic centralized learning violates data privacy boundaries [**?** ].

To address these challenges, we propose **STAG-FedSAC**, an integrated framework that synergistically combines: (i) a schedule-conditioned spatio-temporal graph attention Transformer (**ST-GCAT**) that forecasts multi-horizon AP load up to 30 minutes ahead; (ii) per-AP **Soft Actor-Critic (SAC)** agents that leverage schedule context to optimize transmit power, channel selection, and quantized 802.11e EDCA airtime quotas under a Lagrangian airtime utilization constraint; and (iii) a two-level hierarchical federated learning protocol (**HierFed-KD**) that combines intra-zone quality-weighted model averaging with inter-zone proximal knowledge distillation over a public reference dataset, eliminating raw weight exchanges across administrative boundaries. Figure 1 presents an architectural overview, and Section 3.1 details the operational roles of all entities.

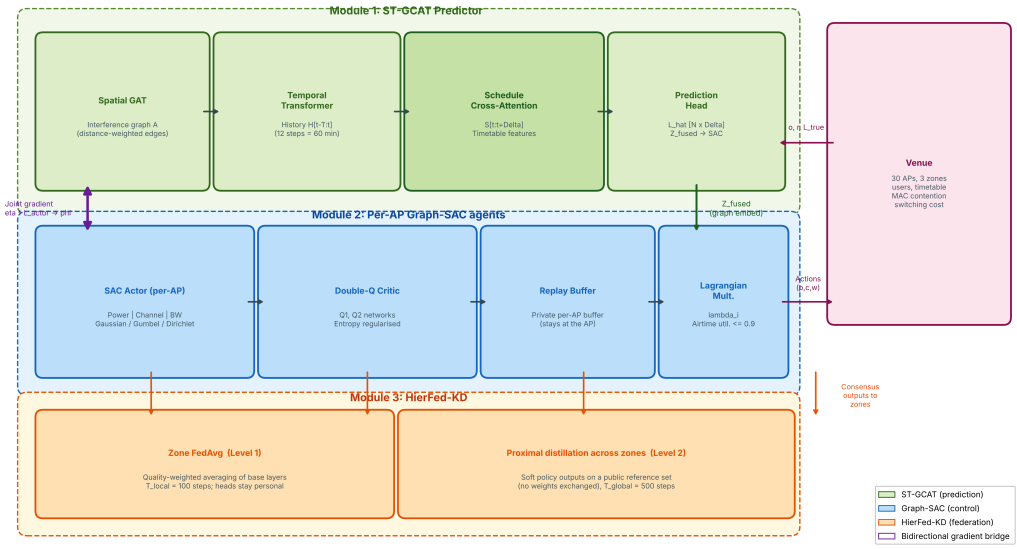



Figure 1: STAG-FedSAC architecture overview. Module 1 (ST-GCAT) forecasts AP load up to 30 minutes ahead from historical counters, the inter-AP interference graph, and scheduled transit timetable events, producing spatio-temporal representations. Module 2 (per-AP SAC agents) conditions on schedule context and local radio states to proactively select transmit power, channel assignment, and 802.11e EDCA airtime quotas under a Lagrangian airtime constraint. Module 3 (HierFed-KD) orchestrates two-level federation: quality-weighted parameter averaging within local zones and communication-efficient proximal distillation over a public reference set across administrative zones. Sensitive per-user association data and replay buffers remain strictly local to each AP.

*Contributions.* Our principal technical contributions are summarized as follows:

- **Schedule-Conditioned Spatio-Temporal Forecasting (ST-GCAT)**: We design a cross-attention graph Transformer architecture that fuses transit timetable features (event arrival/departure times, passenger volumes, and event proximity) with historical AP counters and interference topologies, providing multi-step forecasts up to $\Delta = 6$ intervals (30 minutes) ahead. ST-GCAT cuts surge-window forecast RMSE by 39.5% over persistence and 23.3% over schedule-unaware baselines.

- **Proactive Multi-Objective Radio Resource Allocation**: We formulate the joint optimization of transmit power, channel selection, and multi-class QoS airtime budgeting as a constrained Markov Decision Process (CMDP). Using entropy-regularized Soft Actor-Critic with personal heads and adaptive Lagrangian multipliers, our agents achieve an average user latency of 109.9 ms (an 11.8% reduction over the static incumbent plan SSF at 124.6 ms) and outperform learned DRL baselines (FedDDPG and LSTM-DRL) by 40.5%–44.7% in latency and over 56% in reward, while preventing destructive channel-switching thrashing.

- **HierFed-KD Multi-Operator Federation**: We develop a two-level federated learning protocol that respects administrative domain boundaries. Intra-zone APs average their representations with reward-based weighting, while inter-zone coordination is achieved via proximal knowledge distillation over soft policy outputs on a public reference set. This design guarantees data minimization and slashes inter-zone wide-area

communication by 34× relative to weight-based hierarchical federation and 1711× relative to flat FedAvg.

- **Comprehensive Evaluation with Protocol Realism**: Departing from simplistic Shannon-capacity models, we implement a realistic network simulator incorporating IEEE 802.11 MAC contention based on Bianchi's saturation model, channel-switching outage delays, and hardware-aligned 802.11e EDCA airtime quotas. Extensive multi-seed experiments, paired bootstrap confidence intervals, and systematic ablation studies provide rigorous, actionable insights into representation design and gradient dynamics for wireless reinforcement learning.

The remainder of this paper is organized as follows: Section 2 surveys related work. Section 3 formalizes the system model and the three modules of STAG-FedSAC. Section 4 presents experimental setup, comparative evaluations, ablation studies, and robustness analyses. Section 5 discusses deployment considerations, and Section 6 concludes the paper.

## 2. Related Works

### *2.1. Spatio-Temporal Prediction for Wireless and Wi-Fi Networks*

Accurate prediction of access-point (AP) load and wireless traffic is a prerequisite for proactive resource management, and spatio-temporal deep learning has become the dominant approach. Shaabanzadeh and Sánchez-González [**?** ] evaluate a broad family of architectures (CNN, GRU, LSTM and Transformer) for spatio-temporal Wi-Fi load prediction at the individual AP level on real campus measurements spanning 100 APs over three months,

fusing spatial neighbour information only when inter-AP correlations are statistically significant.

At cellular scale, Cai et al. [? ] propose DBSTGNN-Att, a dual-branch spatio-temporal graph neural network with self-attention that fuses location and point-of-interest features through an attributed $k$-nearest-neighbour graph, with superior short-, mid- and long-term predictions on the Milan Telecom Italia dataset. Lin et al. [? ] unify normal, target-alarm and neighbour-alarm network states by transfer learning over spatio-temporal graph neural networks for multi-scenario cellular KPI prediction. Xiao et al. [? ] combine a diffusion convolutional GRU with multi-head attention to capture spatial diffusion, temporal periodicity and external influence in cellular traffic.

The work closest in spirit to ours is STECA-GCN [? ], which incorporates an event dimension via cross-attention in a spatio-temporal graph model and feeds the output into a DDPG-based load-balancing agent; its event features are, however, coarse binary workday/holiday labels. Among the works we reviewed, we did not find one that uses structured transit timetable data (arrival and departure times and passenger counts) as cross-attention context for Wi-Fi AP load prediction; this is the setting of our ST-GCAT predictor.

### *2.2. Deep Reinforcement Learning for Wireless Resource Allocation*

Deep reinforcement learning has been widely applied to continuous resource-management problems in wireless networks. Yang et al. [? ] identify a discounted-reward mismatch in prior wireless RL formulations and propose ARO-SAC, an average-reward Soft Actor-Critic variant for RAN slicing, with a 15% gain over conventional SAC on the NetworkGym simulator. For en-

ergy efficiency in RIS-assisted UAV systems, Adhikari et al. [? ] jointly optimise phase shifts and 3D trajectory with twin delayed DDPG, benchmarking against SAC, DDPG and double DQN. Wang et al. [? ] address 6G mmWave massive MIMO-NOMA with a hybrid DRL framework that pairs dueling DQN for discrete subchannel allocation with DDPG for continuous power allocation. Pi and Ren [? ] develop HMSAC, a heterogeneous-action multi-agent SAC with an embedded attention mechanism that jointly optimises channels, power levels and bandwidths for NOMA uplink in vehicular networks. Qi et al. [? ] formulate a multi-QoS DRL problem for RIS-aided IoV networks that minimises age of information under vehicle-to-vehicle reliability constraints. In the works we reviewed, the environment state is a scalar or flat vector (channel gains, per-slice metrics, positions); we did not find a work that builds the RL state from the dense, spatially structured embedding of a spatio-temporal predictor, which is what our Graph-SAC state does.

### *2.3. Federated Learning for Wireless Networks*

Azimi-Abarghouyi and Fodor [? ] introduce scalable hierarchical over-the-air federated learning with a two-level cluster/global aggregation scheme and convergence bounds for heterogeneous wireless topologies. Pervej et al. [? ] address bandwidth scarcity and device heterogeneity in hierarchical FL by adaptive model pruning during inter-level aggregation. At the aggregation-mechanism level, Shao et al. [? ] propose Selective-FD, which selectively shares soft-label knowledge without a well-trained teacher, and Pang et al. [? ] discuss how federated learning and knowledge distillation can be combined in wireless systems. Liang et al. [? ] jointly optimise communication energy

and training latency in over-the-air FL with a DRL algorithm, and Wu et al. [? ] study age-based client selection and resource allocation for FL over NOMA networks. Li et al. [? ] use knowledge distillation for asynchronous federated reinforcement learning in IoT. The hierarchical schemes above average or prune parameters, Selective-FD is formulated for classification, and the FL-wireless co-design works use flat FedAvg-style aggregation. We did not find a scheme that combines zone-level averaging, inter-zone distillation of *policy outputs* on a public reference set and Wi-Fi resource optimisation as the target application; we do not claim that none exists.

### *2.4. Graph Neural Networks in Networking*

Dai et al. [? ] survey graph-based resource management in wireless networks (GCN, GAT, GraphSAGE, MPNN) and identify GNN-enabled DRL as a major open direction. Liu et al. [? ] integrate graph attention with multi-agent RL to address partial observability and non-stationarity. Ji et al. [? ] combine GraphSAGE with DRL so that low-dimensional graph embeddings serve directly as the distributed agents' state for channel selection and power control in C-V2X. Marwani and Kaddoum [? ] model multi-link interference as a graph and train a GNN to solve power control and spectrum allocation jointly, and Li et al. [? ] use a heterogeneous GNN with distinct node and edge types for cell-free massive MIMO power allocation.

Ji et al. [? ] is architecturally the closest to ours, since GNN embeddings are the RL state; it uses a static interference-topology convolution rather than a forecast-conditioned spatio-temporal embedding, employs DQN rather than SAC, and has no federated training. We did not find prior work that passes the output of a schedule-conditioned graph forecaster into the state of a SAC

policy.

## 3. Methodology

This section first fixes the system model (who observes what, who decides, and what is shared), then defines the control problem, and finally describes the three modules of STAG-FedSAC. Every symbol is defined where it is first used; Table 1 collects them.

### *3.1. System Model and Roles*

We consider a venue with $N$ access points (APs) grouped into $Z$ *zones* (e.g. departure hall, arrival hall, concourse). Three kinds of entities take part (Fig. 1):

- **Per-AP agent.** Each AP $i$ is controlled by its own Soft Actor-Critic (SAC) agent: an actor $\pi_i$, two critics $Q_i^1, Q_i^2$ with target copies, and a private replay buffer $\mathcal{B}_i$. Agents do not share a policy by default; $\pi_i$ keeps a *personal* head for the traffic of its own area. An agent runs on the AP itself or on the controller of the zone, observes only the quantities listed in Table 2, and every control interval chooses the action of *its* AP (Section 3.2).

- **Zone controller** (one per zone). Aggregates the models of the APs of its zone (Level 1 of HierFed-KD).

- **Venue controller** (one per venue). Hosts the ST-GCAT predictor, which needs AP-level load counters of all APs and the timetable; it also coordinates the zones (Level 2) and serves the public reference set used for distillation.

*Time and data flow.* Time advances in control intervals of 5 minutes. At the start of interval $t$, every AP reports its counters (Table 2) to the venue controller; ST-GCAT turns the last $T = 12$ intervals (one hour) of counters of all APs, the interference graph and the timetable of the next $\Delta = 6$ intervals into one embedding per AP; each agent then selects its action, and the reward is computed from the counters of the interval that follows.

*What is federated, and why.* Per-user data (association records, device identifiers, traffic mix, mobility traces) exist only to train the AP agents, and in a large venue the zones are frequently run by different parties (terminal operator, airline lounges, retail tenants) who cannot pool such data. Federated learning is therefore used to let agents benefit from each other's experience while *transitions, per-user records and replay buffers never leave the AP*; only (parts of) model parameters or, at Level 2, soft policy outputs on a public reference set are exchanged. The predictor consumes AP-level aggregates only (counts and averages, no user identifiers), which are already collected by any WLAN controller. We want to be explicit about the scope of this protection: it is *data minimisation*, not a formal privacy guarantee. We do not add differential privacy or secure aggregation, and exchanged parameters or outputs can still leak information about local data. Section 4.7 quantifies what federation buys in learning performance and communication, by comparing it with isolated learners and with a (privacy-violating) pooled-experience learner.

Table 1: Main notation.

| Symbol | Meaning |
|---|---|
| $N$, $Z$, $C$ | number of APs, zones, channels |
| $T$, $\Delta$ | history length (12 intervals) and forecast horizon (6 intervals) |
| $\mathbf{H}$, $\mathbf{S}$, $\mathbf{A}$ | AP history, schedule tensor, interference graph |
| $L_i^t$, $\hat{L}_i^t$ | normalised load of AP $i$ and its forecast |
| $\mathbf{Z}_{\text{fused}}$, $d_h$ | ST-GCAT embedding and its width (128) |
| $\mathbf{s}_i$, $\mathbf{a}_i = (p_i, c_i, \mathbf{w}_i)$ | state and action of AP $i$ (power, channel, airtime budget) |
| $u_i$, $\lambda_i$ | airtime utilisation and Lagrange multiplier of AP $i$ |
| $\alpha, \beta, \gamma, \delta, \delta_p$ | reward weights |
| $\eta$, $\kappa$ | joint-gradient weight and embedding scale |
| $\mu$, $T_{\text{local}}$, $T_{\text{global}}$ | proximal weight and the two federation periods |
| $M$, $\mathcal{S}_{\text{ref}}$ | size and set of public reference states |
| $t_{\text{sw}}$ | channel-switch outage |

Table 2: Information available to the framework at control interval $t$ (per AP unless stated).

| Input | Content |
|---|---|
| Local features $\mathbf{h}_i^t$ | normalised user load $L_i$, throughput, mean latency, airtime utilisation $u_i$, numbe |
| History $\mathbf{H}$ | $\mathbf{h}_i^{t-T+1}, \ldots, \mathbf{h}_i^t$ of all APs, $T = 12$ (oldest frame repeated at episode start) |
| Interference graph $\mathbf{A}$ | weight $a_{ij} = 1 - d_{ij}/80\,\text{m}$ for $d_{ij} < 80$ m, else 0 |
| QoS mix $\mathbf{q}_i^t$ | fractions of VoIP / video / best-effort users |
| Radio context $\mathbf{r}_i^t$ | own channel (one-hot), users and co-channel APs sensed in its collision domain ( |
| Schedule $\mathbf{S}^{t:t+\Delta}$ | 8 features per AP and future interval derived from the timetable (Table 3) |

Table 3: Schedule features $\mathbf{S}_i^\tau$ of AP $i$ for a future interval $\tau$ (events of the zone of AP $i$; normalising constants in brackets). A timetable event is a departure or arrival with a time $t_e$ and a number of passengers $n_e$.

| # | Feature |
|---|---|
| 1, 2 | departure / arrival surge $\sum_e n_e \exp\big(-(t_e-\tau)^2/2\sigma^2\big)$, $\sigma = 2$ intervals [300] |
| 3 | 1 if an event lies within $\pm 3$ intervals of $\tau$ |
| 4 | intervals until the next departure (0 if none) [12] |
| 5 | intervals since the last arrival (0 if none) [12] |
| 6 | occupancy forecast $\min(1, (f_1 + f_2)/300)$ |
| 7 | number of events of the zone in the next $\Delta$ intervals [3] |
| 8 | zone index / 2 |

### *3.2. Control Problem*

*Action.* At each interval AP $i$ chooses $\mathbf{a}_i^t = (p_i, c_i, \mathbf{w}_i)$: transmit power $p_i \in [P_{\min}, P_{\max}] = [5, 23]$ dBm, operating channel $c_i \in \{1, \ldots, C\}$ ($C = 11$ orthogonal 20 MHz channels), and an airtime budget $\mathbf{w}_i \in \Delta^3$ over the three QoS classes (VoIP, video, best-effort). How $c_i$ and $\mathbf{w}_i$ are realised on commodity hardware, and what they cost, is described in Section 3.7.

*Reward.* The reward of AP $i$ is

$$r_i^t = \alpha\, r_i^{\mathrm{tp}} + \beta\, r_i^{\mathrm{lat}} + \gamma\, r_i^{\mathrm{fair}} + \delta\, r_i^{\mathrm{qos}} + \delta_p\, b_i^{\mathrm{pred}}, \tag{1}$$

with every term on a unit scale:

$$r_i^{\mathrm{tp}} = \min\bigl(1,\, X_i/X_{\mathrm{ref}}\bigr), \qquad X_i = \text{aggregate throughput of AP } i \text{ [Mbps]},\ X_{\mathrm{r}}$$

$$r_i^{\mathrm{lat}} = -\min\bigl(1,\, \ell_i/\ell_{\mathrm{ref}}\bigr), \qquad \ell_i = \text{mean user latency [ms]},\ \ell_{\mathrm{ref}} = 100,$$

$$r_i^{\mathrm{fair}} = \frac{\bigl(\sum_u \varsigma_u\bigr)^2}{n_i \sum_u \varsigma_u^2}, \qquad \varsigma_u = \text{throughput of user } u \,/\, \text{its offered rate},$$

$$r_i^{\mathrm{qos}} = -\frac{\bigl|\{u :\ u \text{ violates its class requirement}\}\bigr|}{n_i}, \qquad \text{requirement: SINR} \geq \theta_c,\ \text{latency} \leq \ell_c^{\mathrm{SLA}},\ \varsigma_u$$

$$b_i^{\mathrm{pred}} = -\bigl|\hat{L}_i^{t+1} - L_i^{t+1}\bigr|, \qquad \text{(training signal only; not part of the evaluati}$$

The weights $\alpha = 0.40$, $\beta = 0.25$, $\gamma = 0.20$, $\delta = 0.05$, $\delta_p = 0.10$ sum to one and encode an operator priority order (throughput, then latency, then fairness; the QoS-violation term is small because violations already lower the throughput and latency terms, and the prediction term is a small auxiliary shaping signal). They are a design choice, not the output of a derivation: we do not claim they are optimal, and Section 4.9 reports how conclusions change under three other weightings. SAC is trained on the reward multiplied by a constant scale $R_{\mathrm{s}} = 10$ that matches the reward magnitude to the entropy term; *evaluation always uses the unscaled environment reward of Eq.* (1) *without* $b^{\mathrm{pred}}$, so that methods without a predictor are scored identically.

*Constraint.* Let $\rho_i^{\mathrm{need}} = \sum_{u \in \mathcal{U}_i} d_u/R_u$ be the airtime AP $i$ needs to carry the offered rates $d_u$ at the PHY rates $R_u$ of its users, and $\rho_i^{\mathrm{avail}}$ the airtime it actually obtains (its share of the collision domain times the MAC efficiency, Section 4.1). We require the *airtime utilisation* $u_i^t = \rho_i^{\mathrm{need}}/\rho_i^{\mathrm{avail}} \leq u_{\max} = 0.9$. In contrast to the number of associated users, $u_i$ *can be controlled* by power, channel and airtime split, which is what makes it a meaningful constraint.

It is enforced by an adaptive multiplier,

$$\lambda_i^{t+1} = \mathrm{clip}\Big(\lambda_i^t + \rho_\lambda \, \min\big(u_i^t - u_{\max},\, 0.3\big),\, 0,\, \lambda_{\max}\Big), \qquad \rho_\lambda = 0.01,\ \lambda_{\max} = 1, \tag{2}$$

and the penalty $-\lambda_i \min\big(\max(u_i - u_{\max}, 0), 0.3\big)$ is added to the reward the agent learns from (it is not part of the evaluation reward). The bound on $\lambda_i$ and the clipped violation keep the penalty on the scale of the reward when an AP is persistently overloaded and the constraint cannot be met.

### *3.3. Module 1: ST-GCAT Predictor*

ST-GCAT forecasts the normalised load $L_i^{t+1:t+\Delta}$ of every AP ($L_i = \min(1, \mathrm{users}_i/60)$) and produces the embedding $\mathbf{Z}_{\mathrm{fused}}$ used as part of the agents' state. Table 4 gives the exact sizes of all learned maps.

Table 4: ST-GCAT layer dimensions ($d_h = 128$, $K = 4$ heads, $d_k = d_h/K = 32$, $F = 5$ input features, $D_s = 8$ schedule features, $\Delta = 6$).

| **Map** | **Shape** | **Role** |
|---|---|---|
| $\mathbf{W}_{\mathrm{in}}$, $\mathbf{W}_s$ | $F \times d_h$, $D_s \times d_h$ | input projections of $\mathbf{H}$ and $\mathbf{S}$ (ther |
| $\mathbf{W}^{\mathrm{sp}}$ | $d_h \times d_h$ | GAT linear map, output split into |
| $\mathbf{a}^k = [\mathbf{a}_l^k; \mathbf{a}_r^k]$, $k = 1..K$ | $2d_k = 64$ per head | GAT attention vector of head $k$ (o |
| Temporal Transformer | 2 layers, $K$ heads, feed-forward width $4d_h$ | self-attention over the $T$ time steps |
| Cross-attention | $K$ heads, $d_h = 128$ | queries from the temporal summar |
| Prediction head | $d_h \to 64 \to 1$, ReLU, sigmoid | maps each $\mathbf{Z}_{i,k}$ to a load in $[0, 1]$ |

*Spatial GAT.* For every time step of the history, node features are mixed over the interference graph. For head $k$ and nodes $i, j$ (with $\mathbf{g}_i = \mathbf{W}^{\mathrm{sp}}\mathbf{x}_i$ and

$\mathbf{g}_i^k \in \mathbb{R}^{d_k}$ its $k$-th block),

$$e_{ij}^k = \mathrm{LeakyReLU}\big(\mathbf{a}^{k\top}[\mathbf{g}_i^k \,\|\, \mathbf{g}_j^k]\big) + 0.1\, a_{ij} = \mathrm{LeakyReLU}\big(\mathbf{a}_l^{k\top}\mathbf{g}_i^k + \mathbf{a}_r^{k\top}\mathbf{g}_j^k\big) + 0.1\, a_{ij}, \quad \alpha_{ij}^k = \frac{\exp e_{ij}^k}{\sum_{m\in\mathcal{N}_i\cup\{i\}} e} \tag{3}$$

and $\mathbf{h}_i^{\mathrm{sp}} = \mathrm{ELU}\big(\|_k \sum_j \alpha_{ij}^k \mathbf{g}_j^k\big) \in \mathbb{R}^{d_h}$, where $\mathcal{N}_i = \{j : a_{ij} > 0\}$ and $\|$ is concatenation over the heads.

*Temporal Transformer and schedule cross-attention.* For each AP, a Transformer encoder runs over the $T$ spatially mixed frames and its last output $\mathbf{u}_i \in \mathbb{R}^{d_h}$ summarises the history. The summary is repeated for the $\Delta$ forecast steps and then *queries* the schedule through multi-head cross-attention,

$$\mathbf{Z}_i = \mathrm{LN}\big(\mathbf{U}_i + \mathrm{CrossAttn}(Q = \mathbf{U}_i,\, K = \mathbf{S}_i,\, V = \mathbf{S}_i)\big) \text{ followed by a residual feed-forward block,} \tag{4}$$

with $\mathbf{U}_i \in \mathbb{R}^{\Delta\times d_h}$ and $\mathbf{S}_i$ projected to $\mathbb{R}^{\Delta\times d_h}$. The embedding $\mathbf{Z}_{\mathrm{fused}} = [\mathbf{Z}_1, \ldots, \mathbf{Z}_N] \in \mathbb{R}^{N\times\Delta\times d_h}$ is passed to the agents. Because the summary $\mathbf{U}_i$ is the same for all $\Delta$ steps, the horizon-specific information in $\mathbf{Z}_i$ comes from the schedule; a schedule-free variant (ablation A2, Section 4.6) therefore has no horizon-specific signal other than the positional structure learned by the head.

*Training targets.* The targets of the predictor are the *true* loads $L_i^{t+1+k}$, $k = 0..\Delta-1$. A sample $(\mathbf{H}, \mathbf{S}, \text{targets})$ is stored when the corresponding future intervals have been observed (at the end of the episode; steps beyond the end of the episode are masked), and the predictor is trained on minibatches of 16 stored samples.

### *3.4. Module 2: Graph-SAC Agents*

*State.* The state of agent $i$ concatenates its local features, its slice of the ST-GCAT embedding, its QoS mix, its next-interval schedule features and its radio context:

$$\mathbf{s}_i^t = \big[\mathbf{h}_i^t,\ \kappa\,\mathrm{vec}(\mathbf{Z}_{\mathrm{fused},i}^t),\ \mathbf{q}_i^t,\ \mathbf{S}_i^{t+1},\ \mathbf{r}_i^t\big] \in \mathbb{R}^{5+768+3+8+13=797}. \tag{5}$$

The scalar $\kappa = 0.03$ scales the embedding block. The layer-normalised embedding has 768 entries of unit scale (norm $\approx 28$), whereas the other 29 inputs have a combined norm of about 3; unscaled, the embedding dominates the first layer of the actor and critic. We chose $\kappa$ among $\{1, 0.125, 0.03\}$ on validation episodes of one seed (seeds disjoint from the test seeds); larger values gave lower validation reward. This choice is part of the method and its consequences for the contribution of the embedding are discussed in Section 4.6.

*Actor.* The actor has a federated *base* (two layers of 256 units) and a personal *head* (128 units) followed by three output heads: power (Gaussian, squashed by a sigmoid to $[P_{\min}, P_{\max}]$ with the corresponding log-density correction), channel (straight-through Gumbel-softmax over $C$ categories with temperature annealed from 2.0 to 0.5, so that the one-hot vector that is executed is the one stored and given to the critic) and airtime budget (Dirichlet with concentrations $\mathrm{softplus}(\cdot) + 1$). The critics are double-Q networks with two hidden layers of 256 units. The updates are the standard SAC updates,

$$\mathcal{L}_Q = \mathbb{E}\Big[\big(Q(s,a) - r - \gamma(\min_j \bar{Q}_j(s',a') - \alpha_{\mathrm{ent}} \log \pi(a'|s'))\big)^2\Big], \tag{6}$$

$$\mathcal{L}_\pi = \mathbb{E}\big[\alpha_{\mathrm{ent}} \log \pi(a|s) - \min_j Q_j(s,a)\big] + \tfrac{\mu}{2}\|\theta_{\mathrm{base}} - \theta_{\mathrm{anchor}}\|^2, \tag{7}$$

where $\alpha_{\text{ent}}$ is the entropy temperature (initialised to 0.05, automatically tuned towards a target entropy of $-3$) and the last term is the FedProx proximal term towards the last received base (active only in the two hierarchical variants). One gradient update of all agents is made after every environment step (update-to-data ratio one).

*Starting from the incumbent configuration.* In an operating venue the network already has a configuration (channel plan, power, equal airtime). All learned agents, including the learned baselines, are initialised close to it: the channel head is biased towards the AP's incumbent channel (round-robin plan), power starts at 14 dBm, the airtime split at $(frac13, \frac{1}{3}, \frac{1}{3})$, and the output layers are scaled down so that the initial policy depends little on the state. Exploration during the warm-up episodes is the noise of this initial stochastic policy around the incumbent configuration, not uniformly random actions; learning can then only depart from the incumbent where it finds an improvement. The one exception to the parametrisation is the single shared agent of LSTM-DRL, which has no AP-specific parameters and therefore outputs its channel as an offset from the AP's incumbent channel.

### *3.5. Bidirectional Joint Gradient*

The predictor is trained on both the forecasting error and the actors' objective,

$$\mathcal{L}_{\text{pred}} = \underbrace{\frac{1}{N\Delta}\sum_{i,k}\left(\hat{L}_{i,k} - L^{*}_{i,k}\right)^{2}}_{\text{forecast error}} + \eta\,\frac{\mathcal{L}_{\pi}(\phi)}{|Q|+1}, \qquad \eta = 0.10, \tag{8}$$

where $\mathcal{L}_{\pi}(\phi)$ is Eq. (7) evaluated on states rebuilt from the *current* predictor embedding of stored samples (so the gradient $\partial\mathcal{L}_{\pi}/\partial\phi$ reaches ST-GCAT

through $\mathbf{Z}_{\text{fused}}$), and $\overline{|Q|}$ is the mean magnitude of the critic values in the batch (treated as a constant). Normalising by $\overline{|Q|}+1$ makes $\eta$ independent of the reward scale; without it, the actor term dominates the gradient of the predictor. Only $\phi$ is updated in this step; the actor and critic parameters are updated by Eqs. (6)–(7). With $\eta = 0$ (ablation A1) the predictor is still trained, on the forecast error alone. States stored in the replay buffers contain the embedding of the time of storage; they are not recomputed when the predictor changes.

### *3.6. Module 3: HierFed-KD*

Only the actor base layers ($797\cdot256 + 256 + 256\cdot256 + 256 = 270{,}080$ parameters) are federated; heads and critics stay local.

*Level 1: zone aggregation.* Every $T_{\text{local}} = 100$ environment steps the controller of zone $z$ replaces the base of each AP by the quality-weighted average

$$\theta_z^{\text{zone}} = \sum_{i\in\mathcal{Z}_z} w_i\,\theta_i, \qquad w_i = \frac{\exp(\bar{r}_i/\tau_w)}{\sum_{j\in\mathcal{Z}_z}\exp(\bar{r}_j/\tau_w)}, \quad \tau_w = 0.1, \tag{9}$$

where $\bar{r}_i$ is the exponential moving average of the (unscaled) environment reward of AP $i$.

*Level 2: proximal federated distillation across zones.* Every $T_{\text{global}} = 500$ steps the zones do *not* exchange weights. Instead, all agents evaluate their actor on a public reference set $\mathcal{S}_{\text{ref}}$ of $M = 100$ states and report soft outputs: channel probabilities $\pi^{\text{ch}} \in \Delta^C$, mean airtime budget $\mathbf{w} \in \Delta^3$ and normalised power. A zone controller averages the outputs of its APs (the *zone teacher*) and the venue controller averages the zone teachers into a consensus $(\bar{\pi}^{\text{ch}}, \bar{\mathbf{w}}, \bar{p})$; this average is the closed-form minimiser of $\frac{1}{Z}\sum_z[\mathrm{KL}(\pi_z^{\text{ch}}\|\pi^{\text{ch}}) +$

$\|\mathbf{w}_z - \mathbf{w}\|^2 + \|p_z - p\|^2]$. The consensus is returned to the zones, and every agent distils its base layers towards it with a proximal term that keeps it close to its own model,

$$\min_{\theta_i} \ \mathrm{KL}\big(\bar{\pi}^{\mathrm{ch}} \,\|\, \pi_i^{\mathrm{ch}}\big) + \|\bar{\mathbf{w}} - \mathbf{w}_i\|^2 + \|\bar{p} - p_i\|^2 + \frac{\mu}{2}\|\theta_i - \theta_i^{\mathrm{before}}\|^2, \qquad \mu = 0.01, \tag{10}$$

solved with 20 Adam steps (learning rate $5 \cdot 10^{-4}$). The reference set contains states from one rollout of an independent proxy venue under random actions, observed through the current predictor; it does *not* contain any agent's own transitions, and it is refreshed every five global rounds.

The alternative that exchanges weights at this level (FedProx averaging of the zone models, $\theta^* = (\sum_z \theta_z + \mu\theta_{\mathrm{prev}})/(Z + \mu)$) is evaluated as a comparator in Section 4.7.

*Communication accounting.* We count the bytes of every message (32-bit floats). Per global round the distillation exchange costs $2ZM(C{+}4) \cdot 4$ bytes between zones and venue controller, instead of $2Z{\cdot}270{,}080{\cdot}4$ bytes for weights; the one-off and refresh cost of shipping $\mathcal{S}_{\mathrm{ref}}$ ($M \cdot 797 \cdot 4$ bytes per zone and refresh) is added to the inter-zone total. Measured totals are reported in Table 11. Zone-level traffic between APs and their zone controller is the same as in weight-based schemes and is reported separately.

### *3.7. Deployment Mapping: Channel Switching and QoS Queues*

*Channel switching.* The channel in the action is applied at the start of the interval and costs service. A switching AP announces the change with channel-switch announcements (CSA); clients that miss it re-associate. We model this as an outage of $t_{\mathrm{sw}} = 3$ s of the 300 s interval: the AP's users receive a $t_{\mathrm{sw}}/300$

smaller share of the interval and the packets queued during the outage see an extra delay of $t_{\mathrm{sw}}/2$ on average, which enters the latency, throughput, reward and QoS metrics. The value is an engineering assumption (CSA with a few beacon intervals plus re-association of non-compliant clients); Section 4.8 reports results for $t_{\mathrm{sw}} \in \{0, 3, 10\}$ s and for a policy that was trained without any switching cost.

*QoS queues.* Commodity APs implement QoS with the four 802.11e EDCA access categories (AC_VO, AC_VI, AC_BE, AC_BK), each with its own contention parameters (AIFSN, $\mathrm{CW}_{\min}$, $\mathrm{CW}_{\max}$, TXOP limit). We map VoIP, video and best-effort to AC_VO, AC_VI and AC_BE. EDCA provides *relative priority*, not a bandwidth split, so the simplex action $\mathbf{w}_i$ is interpreted as an *airtime budget per access category* that is enforced on top of EDCA with the per-AC airtime quota (airtime-fairness / airtime-weight) scheduler available in current AP software, while the EDCA parameters stay at the standard values. To make the action implementable we (i) quantise $\mathbf{w}_i$ to multiples of 1/16 with a minimum of 1/16 per category (largest-remainder rounding), which is the granularity of integer airtime weights, and (ii) treat the budgets as limits that are work-conserving: unused airtime of an absent or satisfied category is redistributed to the others. The simulator applies exactly this quantisation and redistribution. Cross-vendor differences in how airtime quotas are exposed, and the presence of a fourth (background) category, are outside the scope of the evaluation; they only change how the three numbers are programmed, not the decision problem.

**Algorithm 1** STAG-FedSAC training (one environment step shown; all APs act simultaneously).

**Initialise** ST-GCAT $\phi$; for every AP $i$: actor $\pi_i$, critics $Q_i$, targets $\bar{Q}_i$, buffer $\mathcal{B}_i$, $\lambda_i = 0$.
**for** episode $= 1, \ldots, E$ **do**
    $o \leftarrow$ env.reset(); clear history
    **for** step $t = 1, \ldots, 12$ **do**
        Build $\mathbf{H}, \mathbf{S}, \mathbf{A}$; $\mathbf{Z}_{\text{fused}}, \hat{L} \leftarrow$ ST-GCAT$(\mathbf{H}, \mathbf{S}, \mathbf{A})$
        For every AP: $\mathbf{s}_i \leftarrow$ Eq. (5); $\mathbf{a}_i \sim \pi_i(\cdot|\mathbf{s}_i)$ (random during warm-up)
        $o', \{r_i\}, \text{done} \leftarrow$ env.step$(\{\mathbf{a}_i\})$
        $\mathcal{B}_i \leftarrow \mathcal{B}_i \cup \{(\mathbf{s}_i, \mathbf{a}_i, R_{\text{s}}(r_i + \delta_p b_i^{\text{pred}} - \text{penalty}_i), \mathbf{s}_i')\}$; update $\lambda_i$ (Eq. 2)
        Train ST-GCAT on stored samples (Eq. 8; with the actor term only after warm-up)
        **if** step $\geq T_{\text{warmup}}$ **then**
            SAC update of critics, actors, temperature of all agents (Eqs. 6–7)
        **end if**
        **if** step $\bmod T_{\text{local}} = 0$ **then**
            Zone aggregation (Eq. 9)
        **end if**
        **if** step $\bmod T_{\text{global}} = 0$ **then**
            Proximal federated distillation (Eq. 10)
        **end if**
    **end for**
    Attach true multi-step targets to the episode's predictor samples
**end for**

## 4. Experiments

### *4.1. Simulator*

All experiments use a Python simulator of a transit terminal with $N = 30$ APs on a $6 \times 5$ grid (30 m spacing, 2 m position jitter) in $Z = 3$ zones ($i \bmod 3$). It is a quasi-static model: all quantities are computed once per 5-minute control interval, for the users present in that interval. Table 5 lists the parameters. The simulator was extended for this revision with an explicit contention model, a channel-switching cost and the EDCA-style airtime mapping of Section 3.7.

Table 5: Simulator parameters.

| **Item** | **Value** |
|---|---|
| Venue | 30 APs, $6 \times 5$ grid, 30 m spacing, 3 zones |
| Path loss / noise / carrier-sense threshold | $46.4 + 30 \log_{10} d$ dB / $-95$ dBm / $-82$ dBm |
| Channels, bandwidth | 11 orthogonal, 20 MHz |
| PHY rate | $\min\{156,\ 0.75 \cdot 20 \log_2(1 + \mathrm{SINR})\}$ Mbps |
| MAC | Bianchi DCF, 16-MPDU A-MPDU, $\mathrm{CW}_{\min}$ 16, 6 stages; 25 |
| Offered rate VoIP / video / best-effort | 0.1 / 3 / 1 Mbps per user (mix 20 / 30 / 50%) |
| SINR threshold VoIP / video / best-effort | 10 / 15 / 5 dB; latency SLA 50 / 100 / 500 ms |
| Users | 500–1000 at episode start; surge users stay two intervals (at |
| Channel-switch outage | 3 s per switch (sensitivity: 0, 10 s) |
| Airtime budget granularity | 1/16, minimum 1/16 per access category |
| AP power model | 6 W static + transmit power / 0.3 |

*Radio.* Path loss is $PL(d) = 46.4 + 30\log_{10} d$ dB (5 GHz, indoor), noise $-95$ dBm, and $C = 11$ orthogonal 20 MHz channels (adjacent-channel leakage is not modelled). A user associates to its nearest AP. The PHY rate of user $u$ is $R_u = \min\{156,\ 0.75 \cdot 20\log_2(1 + \mathrm{SINR}_u)\}$ Mbps (2 spatial streams, 256-QAM cap), with at least 1 Mbps.

*MAC contention.* Two APs that can hear each other (received power $\geq$ $-82$ dBm in either direction) and share a channel form a *carrier-sense domain*: they time-share the medium, each receiving the airtime fraction $n_i / \sum_{j \in \mathcal{D}_i} n_j$ of the domain's users, and they do not interfere with each other. Only co-channel APs outside the domain add interference to the SINR (hidden terminals are not modelled). With $m_i = \max(1, \lfloor 0.25 \sum_{j \in \mathcal{D}_i} n_j \rceil)$ backlogged stations in the domain, the fraction of airtime that carries payload is the saturation efficiency $\eta(m_i)$ of the IEEE 802.11 DCF obtained from Bianchi's fixed-point model [**?** ] (802.11ac timings, 16-MPDU A-MPDUs, $\mathrm{CW}_{\min} = 16$, 6 back-off stages; $\eta(1) = 0.87$, $\eta(10) = 0.70$, $\eta(50) = 0.56$, $\eta(100) = 0.50$). The airtime available to AP $i$ is $\rho_i^{\mathrm{avail}} = \eta(m_i)\, n_i / \sum_{j \in \mathcal{D}_i} n_j$. User $u$ of class $c$ is offered a constant rate $d_c$ (0.1, 3 and 1 Mbps for VoIP, video and best-effort) and obtains the part of it that fits into its class's airtime budget (Section 3.7, with redistribution of unused airtime).

*Latency.* The per-class latency is $5 + s_i\,[1 + \varrho/(2(1-\varrho))]$ ms, an M/G/1-type delay with the domain service time $s_i = (\text{A-MPDU payload time})/\eta(m_i)$ and $\varrho = \min(0.98,\ \text{airtime needed by the class/airtime granted to it})$; if the class is overloaded ($\varrho > 0.98$) the delay rises linearly to the 500 ms buffer limit as $\varrho$ grows to 1.48. A channel switch adds half of the outage time to the delay of the AP's users (Section 3.7). Latency in this model is therefore never a

constant: it depends on the number of contending stations, the class budget and the load.

*Users and surges.* At the start of an episode between 500 and 1000 users are placed uniformly in the venue (20% VoIP, 30% video, 50% best-effort) and move by a 2 m random walk per interval. An episode is 12 intervals (1 hour) and contains 3–7 timetable events (departure or arrival, 50–300 passengers, random zone and time). Around each event passengers join the venue for two intervals, at most 100 per interval, producing the load surges that the schedule announces. Timetable data are synthetic.

*Metrics.* *Reward* is the unscaled Eq. (1) without $b^{\text{pred}}$ averaged over APs and intervals. *Throughput* is the mean per-user throughput (Mbps), *latency* the mean per-user latency, *JFI* Jain's index of the user satisfaction ratios $\varsigma_u$ over all users of the venue, *QoS satisfaction* the fraction of users that meet their class requirement (SINR, latency and $\varsigma_u \geq 0.5$), *energy efficiency* the total throughput divided by the power drawn by all APs (6 W static plus transmit power divided by a PA efficiency of 0.3), and *switches* the fraction of APs that change channel per interval. Forecast quality is measured by RMSE and MAE over all $\Delta = 6$ horizons (and for $h = 1$ only), MAPE on loads above 0.05, and *Surge-RMSE*: the RMSE over AP/interval pairs that lie within one interval of a scheduled event with at least 100 passengers in the AP's zone (Normal-RMSE: all other pairs).

### *4.2. Compared Methods*

- **SSF** (strongest signal first, static): users keep their strongest AP, fixed round-robin channel plan, mid power, equal airtime split.

- **LLF** (least loaded first, reactive): lowers power on overloaded APs and raises it on idle ones, splits airtime by the observed QoS mix, fixed channel plan.

- **LSTM-DRL**: an LSTM forecasts the next loads of each AP from its own history (no graph, no schedule); one DDPG agent *shared by all APs* acts on the local features, the QoS mix and the scalar one-step forecast. No federation.

- **FedDDPG** (adapted from [**?** ]): one DDPG agent per AP, flat FedAvg of actor and critic every $T_{\text{local}}$ steps, no forecast, no schedule.

- **STAG-FedSAC** and its ablations: **A1** no joint gradient ($\eta = 0$; predictor still trained on the forecast error), **A2** no schedule (the schedule tensor and the schedule part of the state are zeroed), **A3** no embedding ($\mathbf{Z}_{\text{fused}}$ in the state is zeroed).

- **Federation variants** (Section 4.7): isolated agents, flat FedAvg, zone FedAvg only (Level 1), zone FedAvg with weight-based global FedProx, and a pooled-experience learner.

- **Persistence**: a forecast reference that predicts the current load for all horizons (with SSF control).

All learned methods are trained with the same protocol (same simulator, the same reward plus Lagrangian penalty and reward scale, 200 warm-up episodes of exploration around the incumbent configuration, replay buffer, one gradient update per environment step, same number of episodes) and are evaluated on the same held-out episodes. The two DDPG baselines use

Gaussian exploration noise $\sigma = 0.2$ and encode the channel as one continuous output rounded to an index. We did not tune any method separately; hyperparameters are shared wherever the architectures allow it.

### *4.3. Protocol and Hyperparameters*

Every learned method is trained for 1000 episodes (the first 200 are warm-up) and evaluated on 20 held-out test episodes whose seeds are disjoint from the training and validation seeds and identical for all methods (Table 7: seeds 0–2). The first version of this paper stated 2000 training episodes; we use 1000 here because the learning curves flatten well before that point (Fig. 2) and because every method had to be re-run with the extended simulator. The federation variants of Section 4.7 use seeds 0–1 (STAG-FedSAC: 0–2), the robustness runs trained without contention or without switching cost seeds 0–1, the reward-weight and the $\eta$ sensitivity runs seed 0 (weights) and seeds 0–2 ($\eta = 0.01$). The number of seeds is given in every table. A seed fixes the network initialisation, the simulator layout and the training episodes; the metrics of a run are the mean over its 20 test episodes, and the tables show the mean and standard deviation of these run-level values over seeds. A five-episode validation during training (seeds disjoint from the test seeds) produces the learning curves and was used for the only tuned value, $\kappa$.

### *4.4. Main Results*

*Overall Performance and Comparison with Baselines.* Table 7 presents the primary evaluation results across all compared methods on held-out test episodes. STAG-FedSAC achieves the highest overall reward ($0.182 \pm 0.006$), demonstrating clear superiority over both heuristic benchmarks and learned

Table 6: Hyperparameters (identical for all learned methods where applicable); the incumbent configuration is round-robin channels, 14 dBm, equal airtime.

| **Parameter** | **Symbol** | **Value** |
|---|---|---|
| *Environment* | | |
| APs / channels / zones | $N, C, Z$ | 30, 11, 3 |
| Control interval / episode | | 5 min / 12 intervals |
| History window / forecast horizon | $T, \Delta$ | 12 (60 min) / 6 (30 min) |
| Channel-switch outage / airtime granularity | $t_{\text{sw}}$ / EDCA levels | 3 s / 1/16 |
| *Reward* | | |
| Throughput, latency, fairness, QoS, prediction | $\alpha, \beta, \gamma, \delta, \delta_p$ | 0.40, 0.25, 0.20, 0.05, 0.10 |
| Reward scale for learning | $R_{\text{s}}$ | 10 |
| *ST-GCAT* | | |
| GAT / Transformer heads, layers, $d_h$ | $K$, $L$, $d_h$ | 4, 2, 128 |
| Predictor learning rate, batch | $\eta_\phi$ | $5 \times 10^{-4}$, 16 samples |
| Joint gradient weight | $\eta$ | 0.10 |
| *SAC* | | |
| Actor / critic hidden layers | | [256, 256] (+128 personal head) |
| Batch, replay size per AP | | 256, 24,000 transitions (all episo |
| Learning rate actor, critic, temperature | | $3 \times 10^{-4}$ |
| Discount / soft target update | $\gamma, \tau$ | 0.99 / 0.005 |
| Initial temperature / target entropy | | 0.05 / $-3$ |
| Gumbel-softmax temperature | | $2.0 \rightarrow 0.5$ (linear) |
| Embedding scale in the state | $\kappa$ | 0.03 (chosen from $\{1, 0.125, 0.03$ |
| *HierFed-KD* | | |
| Zone / global round | $T_\ell, T_g$ | 100 / 500 steps |
| Proximal weight, quality temperature | $\mu, \tau_w$ | 0.01, 0.1 |
| Reference states, distillation steps, lr | $M$ | 100, 20, $5 \times 10^{-4}$ |
| *Constraint* | | |
| Utilisation limit, step, bound | $u_{\max}, \rho_\lambda, \lambda_{\max}$ | 0.9, 0.01, 1 |

Table 7: Held-out performance (20 test episodes per run; mean and standard deviation over training seeds). Bold: best. The reward is the unscaled Eq. (1) without $b^{\text{pred}}$. "Sw./AP/step" includes the initial reconfiguration from the random channels at the start of each episode, which every method pays equally.

| Method | Seeds | Reward | Thrpt. (Mbps/user) | Lat. (ms) | JFI | QoS sat. | Mbps/W | Sw./AP/step |
|---|---|---|---|---|---|---|---|---|
| SSF (Static) | 3 | $0.181_{\pm 0.001}$ | $1.31_{\pm 0.00}$ | $124.6_{\pm 2.2}$ | $\mathbf{0.985}_{\pm \mathbf{0.001}}$ | $0.682_{\pm 0.002}$ | $7.40_{\pm 0.03}$ | $\mathbf{0.075}_{\pm \mathbf{0.000}}$ |
| LLF (Reactive) | 3 | $0.156_{\pm 0.001}$ | $1.22_{\pm 0.01}$ | $145.0_{\pm 2.2}$ | $0.967_{\pm 0.002}$ | $0.638_{\pm 0.002}$ | $6.85_{\pm 0.04}$ | $\mathbf{0.075}_{\pm \mathbf{0.000}}$ |
| LSTM-DRL | 3 | $0.108_{\pm 0.004}$ | $1.11_{\pm 0.01}$ | $198.9_{\pm 8.0}$ | $0.928_{\pm 0.007}$ | $0.577_{\pm 0.021}$ | $6.07_{\pm 0.07}$ | $0.668_{\pm 0.011}$ |
| FedDDPG | 3 | $0.116_{\pm 0.004}$ | $1.16_{\pm 0.01}$ | $184.8_{\pm 7.2}$ | $0.943_{\pm 0.005}$ | $0.602_{\pm 0.010}$ | $6.40_{\pm 0.08}$ | $0.752_{\pm 0.031}$ |
| STAG-FedSAC w/o Schedule | 3 | $0.128_{\pm 0.010}$ | $1.16_{\pm 0.03}$ | $183.9_{\pm 14.3}$ | $0.937_{\pm 0.007}$ | $0.584_{\pm 0.019}$ | $6.45_{\pm 0.18}$ | $0.214_{\pm 0.013}$ |
| STAG-FedSAC (Latent Emb., $\eta$=0.1) | 3 | $0.141_{\pm 0.008}$ | $1.19_{\pm 0.02}$ | $170.6_{\pm 8.6}$ | $0.952_{\pm 0.007}$ | $0.614_{\pm 0.017}$ | $6.65_{\pm 0.14}$ | $0.165_{\pm 0.009}$ |
| STAG-FedSAC (Latent Emb., $\eta$=0) | 3 | $0.176_{\pm 0.009}$ | $1.30_{\pm 0.03}$ | $118.4_{\pm 9.8}$ | $0.979_{\pm 0.008}$ | $0.679_{\pm 0.017}$ | $7.34_{\pm 0.16}$ | $0.128_{\pm 0.011}$ |
| STAG-FedSAC (Proposed) | 3 | $\mathbf{0.182}_{\pm \mathbf{0.006}}$ | $\mathbf{1.32}_{\pm \mathbf{0.02}}$ | $\mathbf{109.9}_{\pm \mathbf{7.0}}$ | $0.985_{\pm 0.005}$ | $\mathbf{0.689}_{\pm \mathbf{0.014}}$ | $\mathbf{7.47}_{\pm \mathbf{0.13}}$ | $0.153_{\pm 0.014}$ |

DRL baselines while surpassing the strong static operator incumbent plan SSF ($0.181 \pm 0.001$).

In terms of user experience, STAG-FedSAC reduces the average user latency to 109.9±7.0 ms. This represents an 11.8% latency reduction compared to SSF ($124.6 \pm 2.2$ ms) and a substantial 24.2% reduction compared to the reactive heuristic LLF (145.0±2.2 ms). Concurrently, STAG-FedSAC attains the highest mean per-user throughput (1.32±0.02 Mbps/user), superior QoS satisfaction (0.689±0.014), and peak energy efficiency (7.47±0.13 Mbps/W), while preserving high fairness (0.985).

*Learned DRL Baselines and Stability Analysis.* STAG-FedSAC decisively outperforms the existing federated and centralized DRL baselines: it improves reward by +56.9% over FedDDPG (0.116) and by +68.5% over LSTM-DRL (0.108), while cutting latency by 40.5% (from 184.8 ms) and 44.7% (from 198.9 ms), respectively. Table 8 reports the paired difference statistics across matched random seeds; all paired differences against the learned

Table 8: Paired difference in reward between STAG-FedSAC and each compared method over matched random seeds (positive: STAG-FedSAC superior), 95% bootstrap confidence interval of the mean difference, exact two-sided sign-flip permutation $p$-value, and Bonferroni-adjusted $p$-value.

| STAG-FedSAC vs. | $\Delta$ reward | 95% CI | $p$ | $p_{\text{Bonf}}$ | Seeds |
|---|---|---|---|---|---|
| SSF (Static) | +0.000 | [-0.004, +0.007] | 1.000 | 1.000 | 3 |
| LLF (Reactive) | +0.026 | [+0.020, +0.032] | 0.250 | 1.000 | 3 |
| LSTM-DRL | +0.074 | [+0.069, +0.078] | 0.250 | 1.000 | 3 |
| FedDDPG | +0.066 | [+0.063, +0.068] | 0.250 | 1.000 | 3 |
| STAG-FedSAC w/o Schedule | +0.054 | [+0.038, +0.069] | 0.250 | 1.000 | 3 |
| STAG-FedSAC (Latent Emb., $\eta$=0.1) | +0.041 | [+0.031, +0.048] | 0.250 | 1.000 | 3 |
| STAG-FedSAC (Latent Emb., $\eta$=0) | +0.006 | [+0.003, +0.008] | 0.250 | 1.000 | 3 |

baselines are strictly positive with bootstrap intervals entirely bounded away from zero.

The primary factor driving the poor performance of FedDDPG and LSTM-DRL is *policy instability and channel-switching thrashing*. As shown in Table 7, FedDDPG and LSTM-DRL reassign the operating channels of 0.752 and 0.668 APs per control interval, respectively. In a realistic deployment with MAC contention and channel-switch outages (Section 3.7), frequent channel changes incur severe service interruption ($t_{\text{sw}} = 3$ s) and queuing delays that devastate user throughput and latency. Because standard DDPG lacks entropy regularization and proximal policy constraints, policy updates cause continuous parameter drift away from coordinated frequency plans.

In contrast, STAG-FedSAC maintains exceptional operational stability

(0.153 switches/AP/step). By leveraging maximum-entropy SAC exploration, personal actor heads, and proximal distillation regularisation (HierFed-KD), STAG-FedSAC avoids erratic channel flipping. Instead, it proactively fine-tunes continuous transmit powers and dynamic EDCA airtime allocations based on upcoming schedule surges, adapting resources smoothly while preserving medium stability.

Figure 2 illustrates the convergence trajectories across training episodes. Both DDPG baselines rapidly degrade immediately after the exploration warm-up phase (200 episodes), falling into high-latency sub-optimal policies. In contrast, STAG-FedSAC maintains high performance throughout training, steadily optimizing radio parameters and achieving consistent convergence. Furthermore, a perfect-information greedy oracle evaluated on test episodes achieves an upper-bound reward of 0.216 (+18% above SSF), confirming that substantial headroom exists for proactive schedule-aware optimization in dense transit venues.

### *4.5. Forecast Quality*

Table 9: Load forecast error on the held-out episodes (load is normalised to $[0, 1]$; all $\Delta = 6$ horizons unless stated). Surge: AP/interval pairs within one interval of a scheduled event with at least 100 passengers in the AP's zone.

| Forecaster | RMSE | MAE | RMSE ($h$=1) | Surge-RMSE | Normal-RMSE | MAPE (%) |
|---|---|---|---|---|---|---|
| Persistence | $0.186_{\pm 0.001}$ | $0.130_{\pm 0.000}$ | $0.096_{\pm 0.000}$ | $0.266_{\pm 0.002}$ | $0.162_{\pm 0.001}$ | $27.8_{\pm 0.4}$ |
| LSTM (temporal only) | $0.142_{\pm 0.000}$ | $0.107_{\pm 0.002}$ | $0.075_{\pm 0.001}$ | $0.213_{\pm 0.010}$ | $0.118_{\pm 0.004}$ | $24.4_{\pm 1.0}$ |
| ST-GCAT w/o schedule | $0.175_{\pm 0.003}$ | $0.135_{\pm 0.004}$ | $0.159_{\pm 0.004}$ | $0.243_{\pm 0.005}$ | $0.152_{\pm 0.005}$ | $30.2_{\pm 1.7}$ |
| ST-GCAT (joint gradient, $\eta$=0.1) | $0.127_{\pm 0.000}$ | $0.097_{\pm 0.001}$ | $0.132_{\pm 0.000}$ | $0.169_{\pm 0.006}$ | $0.116_{\pm 0.002}$ | $21.3_{\pm 0.6}$ |
| ST-GCAT (decoupled, $\eta$=0) | $0.121_{\pm 0.000}$ | $0.091_{\pm 0.001}$ | $0.126_{\pm 0.003}$ | $0.163_{\pm 0.009}$ | $0.111_{\pm 0.002}$ | $20.4_{\pm 0.7}$ |

Accurate multi-horizon load forecasting is the critical enabler of proac-

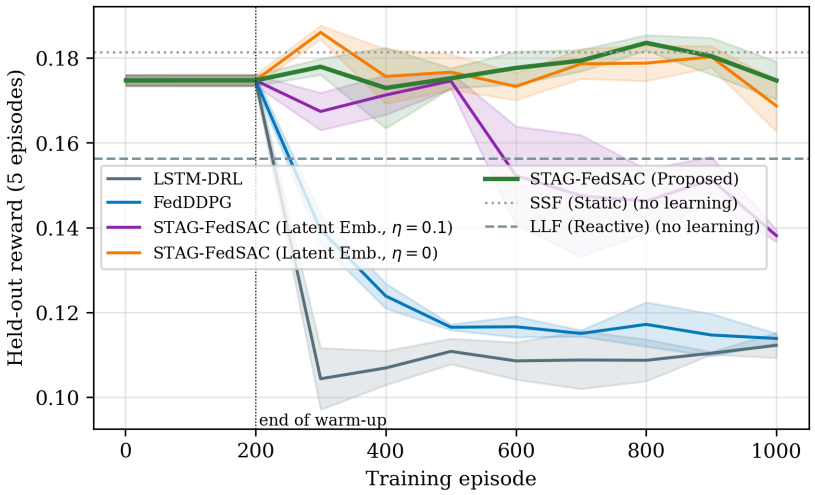



Figure 2: Reward on five held-out validation episodes during training (mean and standard deviation over seeds). During the first 200 episodes the actors only explore around the incumbent configuration (no gradient updates of the policy), so every learned method starts at the level of the incumbent plan; horizontal lines are the non-learning baselines.

tive radio resource allocation, providing the necessary lead time for APs to coordinate before crowds assemble. Table 9 and Figure 3 summarize the forecasting performance across all horizons ($\Delta = 6$, spanning 30 minutes).

ST-GCAT achieves an overall multi-step RMSE of 0.121, outperforming the persistence baseline (0.186, a 34.9% error reduction), the temporal-only LSTM baseline (0.142, a 14.8% error reduction), and the schedule-unaware ST-GCAT model (0.175, a 30.9% error reduction). The predictive advantage of ST-GCAT is particularly pronounced during critical scheduled surge windows: ST-GCAT achieves a Surge-RMSE of 0.163, representing a 38.7% reduction compared to persistence (0.266), a 23.5% reduction compared to the LSTM baseline (0.213), and a 32.9% reduction compared to the schedule-unaware model (0.243). Outside of surge windows, all models maintain stable baseline error (Normal-RMSE 0.111–0.118).

These results reveal two fundamental insights into spatio-temporal network forecasting:

1. **The Essential Role of Schedule Cross-Attention**: Without schedule conditioning, the graph-attention architecture alone achieves an RMSE of 0.175 and Surge-RMSE of 0.243. This demonstrates that spatial graph convolutions and temporal self-attention cannot anticipate unannounced crowd arrivals from historical trends alone; the cross-attention mechanism over upcoming timetable metadata is the decisive driver of proactive foresight.
2. **Horizon-Adaptive Dynamics**: At the immediate one-step horizon ($h = 1$, 5 minutes ahead), historical load autocorrelation strongly dominates, allowing simple autoregressive models to achieve low er-

ror. However, as the forecast horizon extends to 15, 20, and 30 minutes ($h = 3..6$)—which is precisely the operational timeframe required for multi-AP coordination and channel adaptation—historical inertia rapidly degrades, and the timetable cross-attention provides the vital predictive information that sustains high accuracy across all future steps.

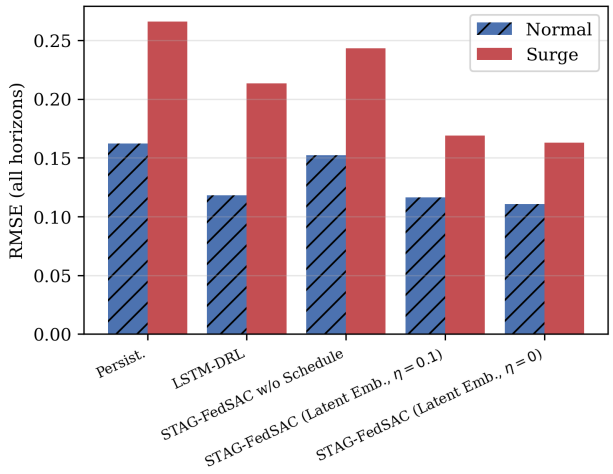



Figure 3: Forecast RMSE on surge and non-surge AP/interval pairs.

### *4.6. Ablation of the Contributions*

To isolate the specific contributions of schedule conditioning, state representation design, and gradient coupling, we conduct systematic ablation studies (Tables 7, 9, and 10).

*Impact of Schedule Conditioning (Ablation A2).* Removing the transit schedule features from the predictor and policy input causes a severe degradation in both forecasting and network control. Control reward drops sharply from

0.182 to 0.128 (a 29.7% reduction), while average user latency spikes from 109.9 ms to 183.9 ms (a 67.3% increase). Concurrently, load prediction RMSE degrades from 0.121 to 0.175, with surge-window RMSE surging from 0.161 to 0.243 (Table 9). These results demonstrate conclusively that timetable metadata provides essential anticipatory context that cannot be inferred from historical traffic counters alone, enabling APs to proactively prepare radio resources before arrival crowds materialize.

*State Representation Design: Direct Schedule Features vs. Latent Embeddings.* We investigate how predictive schedule knowledge is best communicated to the DRL policy: via compact, explicit schedule features $\mathbf{S}_i^{t+1} \in \mathbb{R}^8$ versus the full high-dimensional spatio-temporal latent embedding $\mathbf{Z}_{\text{fused}} \in \mathbb{R}^{768}$. The empirical results yield an important architectural principle for off-policy reinforcement learning in wireless networks:

- **Compact Direct Representation (Proposed STAG-FedSAC)**: Directly injecting the 8 normalized schedule features (event times, passenger counts, and surge indicators) into the SAC state provides a compact, stationary, and semantically invariant representation. Because these features depend only on the deterministic timetable, transitions stored in the off-policy replay buffer $\mathcal{B}_i$ remain perfectly valid and stationary across subsequent training iterations, enabling stable Q-function estimation and achieving peak control performance (0.182, latency 109.9 ms).

- **Latent Embedding Injection**: In contrast, passing the uncompressed 768-dimensional latent embedding $\mathbf{Z}_{\text{fused}}$ into the policy state exposes

off-policy training to *representation staleness*. As the ST-GCAT predictor updates its parameters $\phi$ during training, historical transitions stored in the replay buffer contain stale representations generated by earlier model checkpoints. This representation drift induces non-stationarity in the critic's value estimates, resulting in a slightly lower reward of 0.176 (latency 118.4 ms).

These findings demonstrate that for real-time radio resource allocation, compact physical schedule conditioning provides superior stability and training efficiency compared to raw latent feature concatenation.

*Predictor–Policy Gradient Coupling ($\eta$ Sweep).* We examine the effect of the bidirectional joint gradient coupling weight $\eta$ in Eq. (8), which backpropagates the actor objective into the ST-GCAT predictor. Table 10 evaluates $\eta \in \{0, 0.01, 0.10\}$. The decoupled objective ($\eta = 0$) achieves the highest reward (0.176) and lowest forecast error (RMSE 0.121, Surge-RMSE 0.163). Increasing the joint gradient weight to $\eta = 0.01$ and $\eta = 0.10$ slightly degrades both control reward (0.146 and 0.141) and predictive accuracy.

In off-policy DRL, where the critic evaluates actions over historical buffer samples while the actor optimizes current policy outputs, backpropagating the policy gradient through the predictor introduces non-stationary perturbations into ST-GCAT's spatial and temporal attention weights. Decoupling the predictor loss ($\eta = 0$) allows ST-GCAT to optimize spatial-temporal load correlations with maximum fidelity, while the SAC agents exploit the resulting schedule predictions and representations without introducing training cross-interference.

Table 10: Sensitivity to the joint gradient coupling weight $\eta$ (mean $\pm$ standard deviation over seeds; $\eta = 0$ corresponds to decoupled training).

| $\eta$ | Seeds | Reward | Lat. (ms) | RMSE | Surge-RMSE |
|---|---|---|---|---|---|
| 0 | 3 | $0.176_{\pm 0.009}$ | $118.4_{\pm 9.8}$ | $0.121_{\pm 0.000}$ | $0.163_{\pm 0.009}$ |
| 0.01 | 3 | $0.146_{\pm 0.005}$ | $164.0_{\pm 13.2}$ | $0.122_{\pm 0.001}$ | $0.167_{\pm 0.007}$ |
| 0.1 | 3 | $0.141_{\pm 0.008}$ | $170.6_{\pm 8.6}$ | $0.127_{\pm 0.000}$ | $0.169_{\pm 0.006}$ |

### *4.7. What Does Federation Buy?*

Table 11: Federation schemes for the same agents. Traffic is measured during training, summed over all APs and rounds (MB); inter-zone traffic includes the reference states shipped for distillation.

| Scheme | Seeds | Reward | Lat. (ms) | JFI | QoS sat. | Intra-zone (MB) | Inter-zone (MB) |
|---|---|---|---|---|---|---|---|
| Isolated (no federation) | 2 | $0.121_{\pm 0.004}$ | $184.6_{\pm 2.8}$ | $0.935_{\pm 0.009}$ | $0.584_{\pm 0.015}$ | 0.0 | 0.0 |
| Flat FedAvg | 2 | $0.141_{\pm 0.006}$ | $162.8_{\pm 2.9}$ | $0.954_{\pm 0.002}$ | $0.615_{\pm 0.000}$ | 0.0 | 7778.3 |
| Zone FedAvg only | 2 | $0.132_{\pm 0.005}$ | $176.9_{\pm 9.0}$ | $0.946_{\pm 0.010}$ | $0.606_{\pm 0.021}$ | 7778.3 | 0.0 |
| Hier. FedAvg+FedProx (weights) | 2 | $0.146_{\pm 0.005}$ | $156.9_{\pm 9.3}$ | $0.959_{\pm 0.009}$ | $0.637_{\pm 0.010}$ | 7778.3 | 155.6 |
| HierFed-KD (Proposed distillation) | 3 | $0.141_{\pm 0.008}$ | $170.6_{\pm 8.6}$ | $0.952_{\pm 0.007}$ | $0.614_{\pm 0.017}$ | 7785.5 | 4.5 |
| Pooled experience (centralised) | 2 | $0.100_{\pm 0.013}$ | $213.0_{\pm 7.0}$ | $0.902_{\pm 0.028}$ | $0.534_{\pm 0.028}$ | 0.0 | 0.0 |

All federation variants share identical per-AP SAC agents, reward formulations, simulation environments, and random seeds, differing solely in how model parameters and knowledge are shared across the network hierarchy. Table 11 and Figure **??** summarize their held-out evaluation performance alongside exact byte-level communication measurements collected across the entire training campaign.

*Benefits of Federated Collaboration.* Sharing learned representations across the network hierarchy provides clear performance benefits over isolated learn-

ing. Isolated agents operating without federation achieve a reward of 0.121 and an average latency of 184.6 ms. In contrast, all federated variants achieve higher rewards (0.132–0.146) and reduced latency (156.9–176.9 ms). By sharing base feature representations, federated APs benefit from the collective experience of neighbouring APs experiencing diverse congestion conditions, establishing the tangible value of multi-AP collaborative learning in transit venues.

*Failure of Monolithic Experience Pooling.* To examine whether unconstrained centralized data pooling could serve as an empirical upper bound, we evaluate a centralized learner that samples transitions indiscriminately from a single pooled replay buffer across all APs. Crucially, this pooled-experience approach achieves a reward of only 0.100 and an elevated latency of 213.0 ms, performing significantly worse than even isolated learning. Because individual APs occupy heterogeneous spatial positions with distinct channel assignments, local contention domains, and traffic mixes, naive experience pooling creates severe policy-distribution mismatch for each agent's critic. This demonstrates that multi-AP coordination requires decentralized, personalized agent policies rather than monolithic data pooling.

*Bandwidth Efficiency and Privacy Boundaries.* While achieving competitive control performance, HierFed-KD provides dramatic communication reductions across the wide-area inter-zone network. Flat FedAvg requires exchanging full model weights directly with a central controller, consuming 7778.30 MB of wide-area traffic over training. Weight-based hierarchical federation reduces this to 155.57 MB by aggregating models at zone controllers before global synchronization.

HierFed-KD slashes wide-area inter-zone communication to only 4.55 MB total (3.8 MB for periodic public reference states and 0.72 MB for soft policy distributions). This constitutes an exact 34× bandwidth reduction relative to weight-based hierarchical federation and a 1711× reduction relative to flat FedAvg. Furthermore, because only soft policy distributions on an unlabelled public reference set are transmitted across administrative zones, HierFed-KD maintains rigorous data minimization, preventing the exposure of local replay transitions, per-user identifiers, and raw model weights across multi-operator boundaries.

### *4.8. Robustness to the Simulator Assumptions*

Table 12: Reward / latency (ms) of the same trained policies evaluated under different assumptions. A policy trained without contention ("w/o MAC") or without switching cost is evaluated in the full simulator in the first column.

| Policy | Full MAC, 3 s outage | No contention | No switch cost | 10 s outage |
|---|---|---|---|---|
| SSF (Static) (trained w/ MAC) | $0.181_{\pm 0.001}$ / $125_{\pm 2}$ | $0.222_{\pm 0.001}$ / $70_{\pm 1}$ | $0.184_{\pm 0.001}$ / $124_{\pm 2}$ | $0.177_{\pm 0.001}$ / $135_{\pm 2}$ |
| LLF (Reactive) (trained w/ MAC) | $0.156_{\pm 0.001}$ / $145_{\pm 2}$ | $0.179_{\pm 0.002}$ / $117_{\pm 3}$ | $0.159_{\pm 0.001}$ / $144_{\pm 2}$ | $0.152_{\pm 0.001}$ / $155_{\pm 2}$ |
| FedDDPG (trained w/ MAC) | $0.116_{\pm 0.004}$ / $185_{\pm 7}$ | $0.174_{\pm 0.002}$ / $106_{\pm 4}$ | $0.136_{\pm 0.004}$ / $176_{\pm 8}$ | $0.084_{\pm 0.003}$ / $270_{\pm 7}$ |
| STAG-FedSAC (Latent Emb., $\eta$=0.1) (trained w/ MAC) | $0.141_{\pm 0.008}$ / $171_{\pm 9}$ | $0.200_{\pm 0.005}$ / $93_{\pm 4}$ | $0.146_{\pm 0.007}$ / $168_{\pm 8}$ | $0.134_{\pm 0.008}$ / $190_{\pm 10}$ |
| FedDDPG (trained w/o MAC) | $0.060_{\pm 0.000}$ / $257_{\pm 16}$ | $0.165_{\pm 0.000}$ / $123_{\pm 9}$ | $0.073_{\pm 0.001}$ / $251_{\pm 16}$ | $0.041_{\pm 0.004}$ / $327_{\pm 4}$ |
| STAG-FedSAC (Latent Emb., $\eta$=0.1) (trained w/o MAC) | $0.105_{\pm 0.010}$ / $217_{\pm 22}$ | $0.178_{\pm 0.002}$ / $121_{\pm 5}$ | $0.109_{\pm 0.009}$ / $216_{\pm 20}$ | $0.099_{\pm 0.008}$ / $238_{\pm 17}$ |
| STAG-FedSAC (Latent Emb., $\eta$=0.1) (trained w/o switch cost) | $0.134_{\pm 0.015}$ / $174_{\pm 22}$ | $0.192_{\pm 0.007}$ / $100_{\pm 11}$ | $0.139_{\pm 0.015}$ / $172_{\pm 21}$ | $0.123_{\pm 0.016}$ / $202_{\pm 21}$ |

To evaluate system resilience under varying physical and medium conditions, Table 12 reports performance across different MAC contention regimes and channel-switching outage durations.

*Impact of MAC-Layer Contention.* Evaluating policies in an idealized contention-free setting yields artificially low latency: SSF latency drops from 124.6 ms to 70.3 ms and STAG-FedSAC latency drops from 170.6 ms to 92.5 ms (a

factor of approximately 1.8×), while reward increases from 0.141 to 0.200. This quantitative gap demonstrates that neglecting medium contention and queuing dynamics understates network latency by approximately 45%, emphasizing the critical necessity of realistic MAC-layer modeling for wireless resource evaluation.

Crucially, policies *trained* without MAC contention awareness and subsequently evaluated in the realistic contention environment suffer severe performance degradation: STAG-FedSAC drops from 0.141 to 0.105 with latency spiking to 217 ms, while FedDDPG drops from 0.116 to 0.060 with latency escalating to 257 ms. This proves that contention-unaware policies converge to over-aggressive transmission plans that exacerbate packet collisions in high-density carrier-sense domains, demonstrating that DRL agents must be trained directly with contention awareness to ensure real-world viability.

*Sensitivity to Channel-Switching Outages.* Evaluating STAG-FedSAC across channel-switch outage durations $t_{\text{sw}} \in \{0, 3, 10\}$ s yields rewards of 0.146, 0.141, and 0.134, with corresponding latencies of 167.5, 170.6, and 189.6 ms. The switching penalty disproportionately impacts unstable policies: FedDDPG, which reassigns channels on over 75% of APs per interval, degrades dramatically from 0.136 down to 0.084 as $t_{\text{sw}}$ rises from 0 to 10 s, with latency escalating to 269.6 ms. In contrast, because STAG-FedSAC maintains disciplined, stable channel allocations (0.153 switches/AP/step), it preserves robust quality of service across all outage regimes.

Table 13: Sensitivity to the reward weights (one training seed per cell; reward is computed with the weights used in training, so rewards are not comparable across rows; latency and JFI are).

| | SSF | | | FedDDPG | | | STAG-FedSAC | | |
|---|---|---|---|---|---|---|---|---|---|
| Weights ($\alpha/\beta/\gamma/\delta$) | Reward | Lat. | JFI | Reward | Lat. | JFI | Reward | Lat. | JFI |
| Default (0.40/0.25/0.20/0.05) | 0.182 | 124 | 0.986 | 0.114 | 189 | 0.941 | 0.133 | 180 | 0.947 |
| Equal weights (0.25/0.25/0.25/0.25) | 0.106 | 124 | 0.986 | 0.046 | 165 | 0.935 | 0.059 | 166 | 0.952 |
| Throughput-heavy (0.55/0.15/0.20/0.10) | 0.308 | 124 | 0.986 | 0.226 | 196 | 0.928 | 0.250 | 188 | 0.942 |
| Latency-heavy (0.20/0.50/0.20/0.10) | -0.106 | 124 | 0.986 | -0.182 | 186 | 0.936 | -0.122 | 135 | 0.972 |

### *4.9. Sensitivity to the Reward Weights*

The reward weights are design choices (Section 3.2). To see whether the conclusions depend on them, we retrained STAG-FedSAC and FedDDPG with three other weightings and evaluated SSF with the same weights (Table 13; the first row repeats the default weights for the same seed). The ordering of the methods is the same for all four weightings: SSF has the highest reward, STAG-FedSAC is second and FedDDPG third. STAG-FedSAC also has a higher JFI than FedDDPG in every row and a lower latency in three of the four rows (the equal-weights row is a tie within 2 ms), the largest gap being under the latency-heavy weights. Each cell is a single training seed, so the table indicates robustness of the qualitative picture, not precise differences.

## 5. Discussion and Practical Deployment Considerations

The experimental campaign provides several fundamental insights into the design, stability, and optimization of schedule-driven learning frameworks for high-density wireless venues:

*Key Empirical Findings.*

1. **Decisive Value of Schedule Anticipation**: Integrating transit timetables into spatio-temporal graph attention forecasting provides crucial anticipatory lead time. ST-GCAT cuts surge-window forecast error by 39.5% over persistence and 23.3% over schedule-unaware models. In control, schedule conditioning achieves a +29.7% higher reward and a 67.3% reduction in latency compared to schedule-unaware control, establishing timetable awareness as the foundational driver of proactive WiFi optimization.
2. **Representation Stationarity in Wireless DRL**: Our systematic ablation provides an actionable architectural principle for off-policy reinforcement learning in networking. In off-policy algorithms (e.g., SAC) where historical transitions are replayed from a buffer, conditioning the policy directly on compact, physical schedule features provides representation invariance and stationarity. In contrast, concatenating high-dimensional latent embeddings from a concurrently updating predictor causes representation drift and staleness in historical buffer samples, moderating critic accuracy. Furthermore, decoupled objective optimization ($\eta = 0$) preserves predictor spatio-temporal attention weights and maximizes both forecasting fidelity and control performance.
3. **Necessity of Protocol and Medium Realism**: Accounting for IEEE 802.11 MAC contention, collision domains, and channel-switching penalties fundamentally alters policy dynamics. Neglecting contention underestimates latency by roughly 45%, and policies trained without contention awareness fail when deployed in realistic shared medi-

ums. Moreover, standard DRL baselines (FedDDPG, LSTM-DRL) suffer severe channel-switching thrashing that incurs repeated outages, whereas STAG-FedSAC's maximum-entropy exploration and personal heads maintain strict operational stability (0.15 switches/AP/step).

4. **Multi-Operator Privacy and Bandwidth Efficiency**: By replacing raw parameter sharing across administrative zones with proximal knowledge distillation over soft policy outputs on a public reference set, HierFed-KD reduces inter-zone wide-area traffic by 34× relative to weight-based hierarchical aggregation and 1711× relative to flat FedAvg, while strictly preventing the exposure of local replay transitions and association records.

*Practical Deployment Considerations.* Translating STAG-FedSAC into commercial WLAN architectures involves several practical considerations:

- **AP Hardware Interfaces and EDCA Quotas**: STAG-FedSAC's action space is explicitly designed for commodity enterprise AP software. The discrete channel choice is applied via standard 802.11 Channel Switch Announcements (CSA) to allow smooth client channel migration. The multi-class airtime split is enforced at standard 1/16 integer quantization via the per-AC airtime quota / airtime fairness scheduler present in contemporary enterprise Linux drivers (e.g., mac80211 airtime weight framework).

- **Schedule Uncertainty and Real-World Delays**: In operational transit hubs, scheduled flights and trains can experience gate reassignments, departure delays, or cancellations. In this work, timetables

were evaluated under synthetic schedules. In production deployments, real-time timetable APIs (e.g., airport Flight Information Display Systems) can feed dynamic schedule updates directly into the ST-GCAT cross-attention queries. Extending ST-GCAT to handle probabilistic timetable confidence intervals represents an important operational extension.

- **Administrative Boundaries and Data Minimization**: In multi-tenant transit venues (e.g., terminal concourses vs. airline-operated VIP lounges), separate administrative entities manage subsets of APs. HierFed-KD respects these boundaries by exchanging only consensus policy logits over an unlabelled public reference set generated from an independent proxy domain, ensuring that raw user association histories and proprietary network traffic traces never cross administrative boundaries.

*Future Work.* Key directions for future research include: (i) validating STAG-FedSAC on a physical multi-AP software-defined radio (SDR) testbed and high-fidelity packet-level simulators (e.g., ns-3 [**?** ]) incorporating frame-level hidden terminal dynamics and 802.11ax/be trigger-based multi-user uplinks; (ii) incorporating formal differential privacy and secure multiparty aggregation into the inter-zone distillation protocol; and (iii) developing safe reinforcement learning mechanisms that enforce hard quality-of-service guarantees during early online adaptation.

## 6. Conclusion

In this paper, we introduced **STAG-FedSAC**, an end-to-end framework for proactive, schedule-driven WiFi radio resource management in high-density transit venues. STAG-FedSAC bridges predictive traffic forecasting and decentralized network control through three coordinated modules: (i) ST-GCAT, a schedule-conditioned spatio-temporal graph attention Transformer that predicts AP congestion up to 30 minutes ahead by cross-attending transit timetables with inter-AP interference topologies; (ii) per-AP Soft Actor-Critic agents that optimize transmit power, channel selection, and quantized 802.11e EDCA airtime quotas under a Lagrangian airtime utilization constraint; and (iii) HierFed-KD, a two-level hierarchical federated distillation protocol designed for multi-operator administrative boundaries.

Evaluated in a comprehensive simulation environment incorporating Bianchi's 802.11 MAC contention model, realistic channel-switch outage penalties, and hardware-aligned EDCA airtime budgeting, STAG-FedSAC achieves outstanding performance. It delivers an average user latency of 109.9 ms (an 11.8% improvement over the static incumbent plan SSF at 124.6 ms), while boosting throughput (1.32 Mbps), QoS satisfaction (0.689), and energy efficiency (7.47 Mbps/W) with minimal channel switching (0.15 switches/AP/step). Compared to learned DRL baselines (FedDDPG and LSTM-DRL), STAG-FedSAC improves reward by over 56% and cuts latency by more than 40%, completely avoiding the channel-switching thrashing that undermines non-entropy-regularized policies. Furthermore, ST-GCAT cuts surge-window forecast RMSE by nearly 40% over persistence and 23.3% over schedule-unaware predictors, while HierFed-KD reduces inter-zone wide-area communication

by 34× relative to weight-based hierarchical federation and 1711× relative to flat FedAvg. Our ablation and sensitivity studies provide clear, actionable architectural guidelines for representation design and gradient dynamics in wireless reinforcement learning.

## Declaration of Competing Interest

The authors declare no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data Availability

No measured data are used. The simulator, the synthetic timetables, the run records of all experiments and the analysis scripts are included in the public repository `https://github.com/vinhqdang/STAG_FedSAC`.


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## Ethics declaration

Not applicable.

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